



eMaintenance—Information logistics for maintenance support

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ABSTRACT

Today's providers of maintenance and in-service support related to modern aircraft are facing major challenges. A central problem with Maintenance, Repair and Overhaul (MRO) as well as support of aircraft and other complex technical systems, is to manage the ever-increasing information flow and system complexity. Both military and commercial operators need to reduce aircraft downtime and maintenance man hours. Increased manual information management has the opposite effect, inducing unnecessary cost and affect efficiency negatively. Organizations developing and providing maintenance support products and in-service support does also need to improve the capability to efficiently exploit the increasing amount of digital product information and design data provided together with hardware (HW) and software (SW) products from aircraft manufacturers, sub-system suppliers and Original Equipment Manufacturers (OEM). One way to increase aircraft availability and improve maintenance and support efficiency, is to speed up the turnaround time for scheduled and unscheduled maintenance. The ultimate goal is risk-based utilization and support, where true Condition-Based Maintenance (CBM) is integrated with current operational requirements and available resources in real-time. This aims at the elimination of all preventive scheduled maintenance based on fixed time intervals and execution of only corrective maintenance that has been predicted and turned into scheduled maintenance facilitated by proper support. To address the challenge of information logistics of digital product data and information within maintenance in-service support, providers need to adapt new methodologies and tools that enable full utilization of the advantages of digital product data and information in processes and business models, e.g., Service-Oriented Architecture (SOA). To implement such improved support solutions in a global-support environment, eMaintenance is seen as one important building block. eMaintenance includes monitoring, collection, recording and distribution of real-time system health data, maintenance-generated data as well as other decision and performance-support information to different stakeholders independent of organization or geographical location, 24 h a day, 7 days a week (24/7). eMaintenance has the potential to improve the management and performance of activities related to the whole maintenance process, and thereby improve the dependability, safety and Life Cycle Cost (LCC) of critical systems. This can be realized through a coordinated application of Information and Communication Technology (ICT) throughout the maintenance and support processes, thus integrating Built-in Test (BIT) systems, external tests at different maintenance echelons, technical information, diagnostics, prognostics and other sources of support information.

The purpose of this paper is to present some results from a joint academic and aerospace industry research project, describing requirements and expectations that are important in a global-support environment, and also to propose some central components in an eMaintenance framework that integrates maintenance and ICT perspectives.

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1. Introduction

Aircraft manufacturers, as well as maintenance and in-service support providers, are experiencing ever-increasing customer requirements to increase dependability and decrease Life Support Cost (LSC). To achieve this, a central problem for the industry is to manage the rapidly increasing information flow that follows the development of more complex and technologically advanced

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aircraft systems [1]. Customers are also demanding improved system availability, safety, sustainability, cost-effectiveness, operational flexibility and tailored worldwide support 24 h a day, 7 days a week (24/7). This changing business environment requires new and innovative solutions as support products and services to satisfy the needs of customers and end-users.

Maintenance and support concepts for modern complex technical systems, such as civil airliners and military combat aircraft, can be described as focussing on optimizing two fundamental and interdependent elements. The first element is the aircraft (i.e., the system-of-interest), which should be designed to maximize the inherent availability through proper reliability and maintainability design within available Life Cycle Cost (LCC) constraints. This design should be balanced with the support system, which is the second element. The support system should be designed to provide necessary support during the utilization and support phases of the system-of-interest's life cycle, which is measured through the support system's maintenance support performance. The support system is an enabling system, i.e., a system that complements a system-of-interest during its life-cycle stages [2], but does not necessarily contribute directly to its function during operation [3]. An enabling system provides functions and services to users and other stakeholders to ensure a proper and efficient function of the system-of-interest [2].

On an operational level, end-users and managers utilizing the support system of a modern aircraft are confronted with a multitude of computerized functions and Information and Communication Technology (ICT) solutions. However, today there is little or no integration of functions and services related to the support system, such as technical information (publications), maintenance programmes, maintenance plans, job cards, fault diagnosis support, amendment services, health and usage monitoring and operational feedback [4,5].

At the same time, producers and suppliers of maintenance products and customer support services are facing escalating challenges trying to sustain high quality and increase service levels for increasingly complex technical systems in an environment characterized by multiple products, suppliers and customers with increasingly stringent requirements. This environment is also to a great extent becoming purely digital, i.e., an increasing amount of software (SW) and hardware (HW) products, design data and other information exchange is provided and communicated solely in digital form. Hence, both customers and suppliers are facing increased complexity levels regarding information related to configuration control and change management for both the aircraft and its support system. This high complexity level of information logistics will hamper the operational effectiveness and drive LCC.

Therefore, suppliers need to change methodologies, tools and processes for information logistics to be both more customer-focused and more efficient for internal development and sustainability. New concepts for the application of ICT need to address information quality, lead time, accessibility 24/7, usability and an overall reduction of cost for information logistics related to maintenance and support. Simultaneously, since much of the managed data and information related to military aircraft are sensitive and confidential the aspect of information security is of utmost importance and need to be considered, e.g. through content classification, authentication and authorization. In fact, information security is so fundamental that it can override the functionality of any solution.

Information logistics need to integrate and exploit the use of maintenance data and consider fleet status, flight operations and maintenance sources. This would provide, maintenance planners, operations and others with tailored information for decision support, derived from common data sources [6].

Examples of these trends are manufacturers' airline in flight information system. It can include multifunction e-applications for flight operations, passenger, cabin crew and maintenance features. Capabilities in flight operations include performance calculations, electronic manuals, technical logbook, crew e-mail, graphical weather, and charts and maps. For maintenance it includes technical logbook, manuals, maintenance tools and performance monitoring [7].

Capability also needs to be developed to enable more agile and efficient use of new maintenance and support functions integrated in the aircraft, exploitation of operational feedback, as well as rapid supplier adaptation to continuously changing customer specific requirements on HW and SW products as well as services.

An example of such solutions is again manufacturers' innovative support strategies, where customers pay for a significant portion of purchased services with data collected during operations. Maintenance, Repair and Overhaul (MRO) providers are also required to provide data such as man hours, downtime and reliability data. In return the manufacturer commit to include promotion of services from the MRO providers, information exchange in terms of access to specific information (draft service bulletins, information on market requirements, advance information on major retrofit programs) and management reviews [7].

2. Scope

This paper focuses on a concept for ICT-based products and services for maintenance of modern aircraft. However, it is believed to be extensively applicable to similar challenges regarding maintenance of other complex technical systems, e.g. within the transport, process and power industries, as well as telecom and health care. The paper presents requirements and needs regarding the mentioned products and services that are important for both suppliers and customers (operators) of modern aircraft systems in a global-support environment and proposes some central components of an information logistic concept for maintenance purposes, called eMaintenance Management Framework (eMMF), that integrates maintenance and ICT perspectives to address the challenges presented above.

The discussion of central characteristics of system-of-interest (e.g. aircraft) and enabling systems (e.g. a support system external to the aircraft) uses nomenclature based on established and agreed international standards. Availability is used as a general term referring to availability performance and its influencing factors: reliability performance (the way the aircraft is designed to eliminate the need of maintenance, i.e., reducing or eliminating the probability of loss of required functions); maintainability performance (the way the aircraft is designed to facilitate maintenance, i.e., to retain or restore the aircraft's required functions); as well as maintenance support performance (effectiveness and efficiency of the maintenance organization). See Ref. [8].

3. Maintenance within aviation

Experience from maintenance and support system development and real-life operations [9], combined with expectations for future scenarios, stress the importance of supporting the operational capability and competitiveness through high availability of aircraft. Increasing civil requirements on mobility and availability, and military operations from provisional bases combined with rapidly shifting conditions, increase the importance of information and decision support to personnel that maintain systems in operation. This high impact of integrated logistic support on

aircraft operations and the need to improve it regarding maintenance management, information system and logistic network is also observed in other research [10–13]. The aircraft and its support system's design need to be very structured and controlled, meeting requirements with the overall objective of maximizing the total system effectiveness at the lowest possible LCC, see Fig. 1.

The design of the support system also needs to take into account other requirements, such as operators' mixed aircraft fleet and requests to consider re-use of existing resources and capabilities (e.g. facilities, manpower, competencies, ICT and standards) and future requirements, operational or product development requirements, as well as desired levels of standardization and self-support [7].

New technology and innovation also drive development and create new needs. A military example is the so-called fourth generation combat aircraft, e.g. Dassault Rafale, Eurofighter Typhoon and Saab Gripen. These are system aircraft with a digital infrastructure and fully integrated computer systems that utilize a common database through standardized interfaces [15,16]. The building principle of a system-of-systems supplies an immense potential for functional development, but also an extremely complex aircraft system [9]. The civilian equivalents, such as the Boeing 787 Dreamliner and Airbus A350, have focused on ICT as a central component for more efficient maintenance and support solutions. A central part of this aircraft technology development is advanced Built-in Test (BIT) systems. The aircraft BIT is constantly being developed to improve safety, maintainability, testability and supportability of the aircraft as well as to provide health information. This extended access to qualified data from operation and maintenance provides the technical foundation for condition monitoring, extensive data recording, test, diagnostics, prognostics and decision support for condition-based utilization and support [9]. These technology trends, together with the pervasiveness of ICT in the maintenance system and support processes are also key-enabling factors for implementation of remote support services accessible worldwide and 24/7. Nevertheless, the BIT is itself a development that adds further requirements and puts strain on supporting ICT frameworks, information logistic processes and maintenance needs [17–19].

Thus, information logistics and new ICT-solutions have become a central aspect and common denominator for many of the new challenges regarding system complexity, configuration control, change management and applicability of product information, both for the systems-of-interest and their related support

systems. To meet the increasing demand on availability and to cut costs, maintenance and support actors are becoming increasingly dependent on ICT to provide timely and accurate information. Efficient ICT-solutions have rapidly risen as a key driver for good decision making and effective operations. One example of an area with great improvement potential is technical publications. A large proportion of the information support used by maintainers within aerospace is still paper documents, or 'paper-on-screen' solutions (e.g. paper publication suits published as PDF). This practice prevails despite the fact that new aircraft complexity, integrated digital systems and air-to-ground real-time communication technologies make document-oriented, paper-based approaches increasingly expensive to produce and offer inadequate usability. Hence, unnecessary costs are generated simply because maintainers often lack products and services for information support that are adapted to specific roles and situations, and therefore cannot work as efficiently as possible [4,20].

With reference to the problem and solution domains presented above, the concept of eMaintenance presented in this paper can be defined as a structured and coherent application of ICT throughout the whole life cycle of the support system, coordinated with technical solutions in the aircraft system. The main goal is to create a more effective and efficient maintenance process by optimization of an eMaintenance framework as a part of necessary maintenance support implemented in the enabling systems, and thereby improve overall system effectiveness.

To secure the performance of an aircraft at a reasonable cost, it is vital that the design of maintenance and product support concepts is done correctly right from the design phase [3,21]. The character, scope and allocation of the support needed, must also be influenced by the customers' competencies, as well as the operational environment, but also organizational and cultural issues [3,22]. The whole maintenance process needs to be taken into consideration, including its phases for management, support planning, preparation, execution, assessment and improvement (see Fig. 2). Hence, the requirements analysis must consider a multitude of aspects, facts, needs and properties of the technical system, customer requirements and the context of utilization and support for the given system, supplier requirements and other stakeholders, as well as environment and legislation.

Another aspect is that traditionally both civil and defence aerospace industries have built and sold systems to operators. Today, this approach is changing and the industry also has to provide services that traditionally have been carried out by the operators or other third-party suppliers. The highest degree of commitment is needed when delivering total care, or functional products, where the customers are offered availability performance at a fixed price. One example of this is the requirement of 'Maintenance Free Operating Period' (MFOP) on the military side

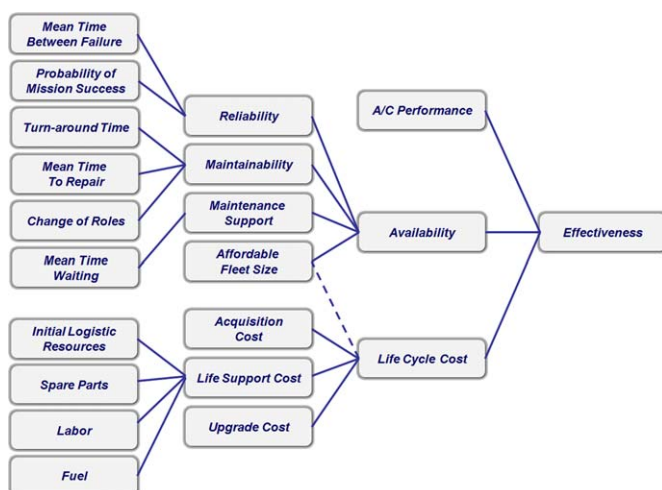


Fig. 1. Important requirements related to system effectiveness [14].

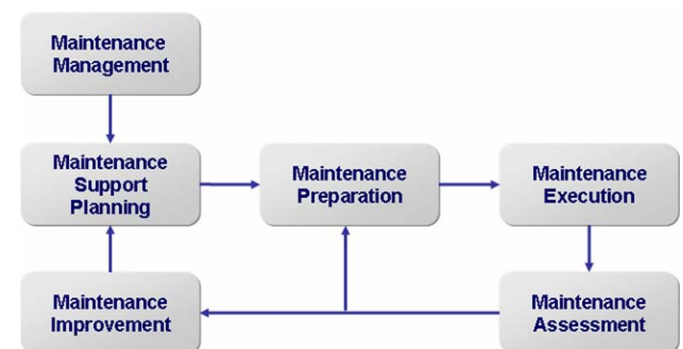


Fig. 2. Phases of an overall maintenance process, adapted from Saab (Internal document) [5].

and 'Power by the Hour' contracts on the civil side. These are market driven requirements on an eMaintenance solution, which introduce additional stakeholder perspectives [20].

4. Maintenance requirements

With reference to the material presented above, the development of products and services integrated with ICT in an eMaintenance solution for support to operation and maintenance of complex systems needs to address a number of central requirements such as

- integration of the system-of-interest and its enabling system to exploit technological development of modern aircraft and ICT in efficient solutions for global maintenance and support 24/7,
- facilitation of configuration control, change management and directive information applicability for both the system-of-interest and products and services included in the enabling system,
- integration of role- and situation-adapted products and services for information and performance support with high accessibility and usability,
- coordination and integration of consumption and production of information at different phases of the maintenance process,
- coordination and integration of tests, diagnostics and prognostics for condition-based utilization and support at different maintenance echelons,
- support to performance-based contracts and functional sales (e.g., power-by-the-hour),
- adherence to international (information) standards.

eMaintenance needs to facilitate support to operators and other stakeholders' work through information and performance support to the overall maintenance process related to modern

aircraft, including ground-support equipment and training devices. eMaintenance also needs to provide a framework and processes for managing product data and information, maintenance and support information, as well as feedback of data and information from operational and maintenance activities.

This spectrum of requirements that built on industrial and operator experience, drives the approach that the improvement of the operative performance and effectiveness of maintenance and support through the introduction of eMaintenance, need to target all phases of the maintenance process (see Fig. 2). In order for the aerospace industry to manage the risks associated with these new, diversified and enhanced solutions in a global-support environment, a wide collection of heterogeneous information sources need to be funnelled and made available for the right user, in the right format, in the right place, at the right time. In the past, the information flow has been quite rigid and inflexible between a rather small numbers of stakeholders. However, in the future the exchange of information is required to reflect closer working relationships and a more dynamic, global-support environment and deliver efficient and tailored support for decision making to stakeholders spanning from OEMs, sub-system suppliers, through system integrators onto operators, maintainers and support providers [20].

Another important requirement is that information logistic concepts within the aerospace industry need to be based on international standards for structured product data sharing. Central among these standards are Product Life Cycle Support (PLCS) [23], Sharable Content Object Reference Model (SCORM) [24] and the AeroSpace and Defence Industries Association of Europe (ASD) specifications suite of standards for Integrated Logistic Support (ILS); S1000D (technical publications), S2000M (material management), the emerging S3000L (Logistics Support Analysis, LSA) and S4000M (Reliability-Centred Maintenance, RCM).

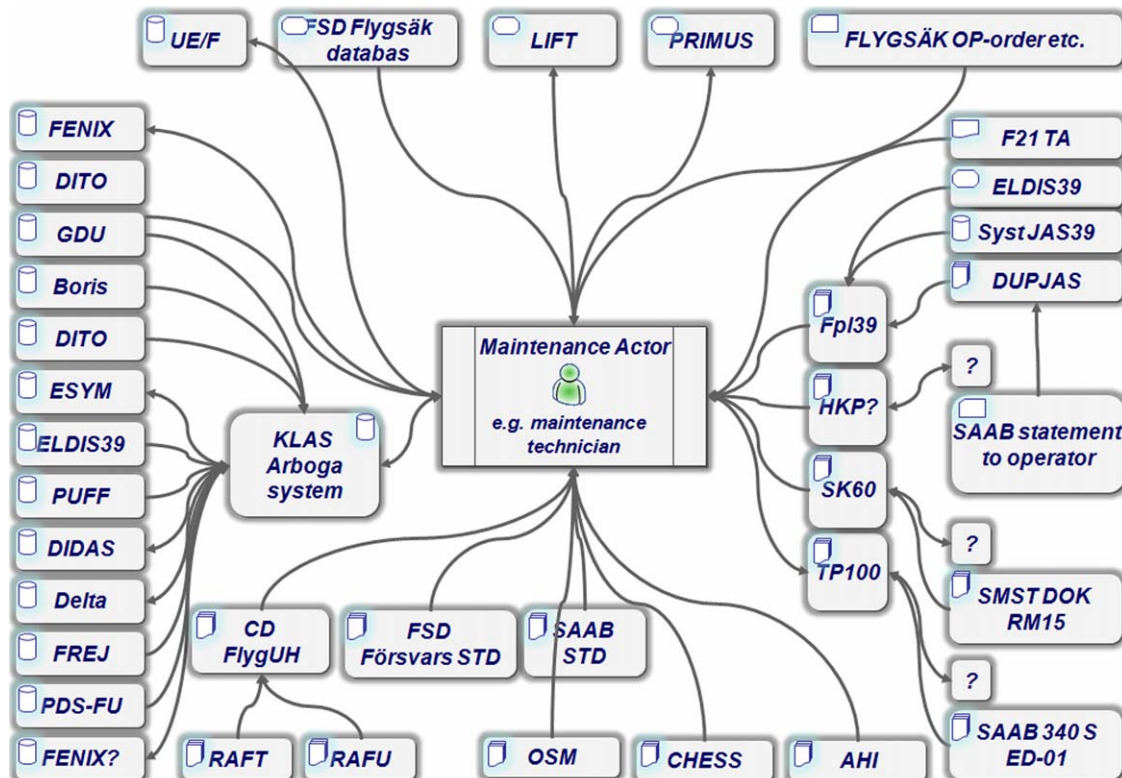


Fig. 3. Illustrating the example of information system needs of a Saab JAS 39 Gripen maintenance technician or planner.

As discussed above, the overall maintenance process involves phases for management, support planning, preparation, execution, assessment and improvement, see Fig. 2. These phases consist of different sets of activities, which are interrelated and adapted to fulfil requirements from different stakeholders. Stakeholders within the maintenance process consume and produce information when performing different activities. For example, during maintenance execution, a maintenance technician receives a work order that requires a maintenance action, and after the performed action the technician reports the outcome. The information from the execution process can further be aggregated to a context in the assessment process. As illustrated in Fig. 3, a maintenance actor, e.g. the maintenance technician, needs to interact with various types and formats of information through different information sources when conducting a maintenance action. This interaction can be reading, writing or both.

Hence, one of the main objectives of establishing an eMaintenance solution is to provide a supporting ICT-environment that improves the performance of the maintenance process by providing seamless and integrated services for sharing content. It is commonly accepted that data collection and distribution is a major issue in the maintenance process and a cornerstone in the area of eMaintenance [5]. It is also accepted that there is a need to be able to convert data to information, and based on this information to generate knowledge valuable for decision making. In order to make the right decisions, there is a need to be able to access and manage the right knowledge. Since knowledge, a term which basically refers to what is known, is not only a summarization of known raw data through information interpretation, but also implies other artefacts, such as documentation and experiences, the focus should be on the content and the content sharing as well as services in the maintenance process rather than on data and data processing [25]. Fig. 4 illustrates information flow between aircraft and some of the related

systems. This information is shared by supplier/provider and customer/operator.

ICT is one of the main prerequisites, not only to improve the effectiveness and efficiency of the maintenance process for complex systems with long life cycles, but also to reduce associated risks, and contribute to a more efficient business process. The utilization of ICT offers a more controlled content sharing, information exchange and knowledge management within the phases of the maintenance process, coordination of the maintenance process with other processes (e.g. operation and modification processes) and a connection to strategic business objectives and external stakeholder requirements [26,27].

5. eMaintenance solutions

In the existing eBusiness environment, information and information systems have become critical maintenance support resources to achieve efficient information logistics.

There are extensive existing efforts that can be adapted to approach issues related to some of these needs. These efforts can be categorized in two groups: (I) generic efforts that are not specifically aimed at maintenance, but might be utilized for maintenance purposes and (II) specific efforts that are specifically aimed at eMaintenance.

Examples of generic efforts are: Essential Unified Process (EssUp), Rational Unified Process (RUP), Service-Oriented Architecture (SOA), Event-Driven Architecture (EDA), Web services, eXtensible Markup Language (XML), XMLSchema, Reference Model for Service-Oriented Architecture, Open Document Format, Microsoft .NET and Biztalk, IBM Web Sphere, Oracle BEA WebLogic and SAP Business suite.

Examples of more maintenance-specific efforts are: Product Life Cycle Support (PLCS), Production Planning and Scheduling (PPS), ISA-95, Open System Architecture for Enterprise Application Integration (OSA-EAI), Common Relational Information Schema (CRIS), condition monitoring and diagnostics of machines-data processing, communication and presentation (ISO 13374), the PROTEUS-platform, the concept of System for Mobile Maintenance (the SMMART-project) with focus on smart tags and wireless communication, tele-maintenance platform (TELMA), Technologies And Techniques for nEW Maintenance concepts (within the TATEM-project) with focus on monitoring technologies related to aviation, Maintenix (MXI Technologies), IFS Applications and SAP Service and Asset Management module within the SAP Business suite.

Some of these efforts contemplate support solutions from a business process perspective (e.g. RUP and EssUP), while others focus on technological aspects of service provision (e.g. OSA-EAI, PROTEUS, DYNAMITE and Maintenix). However, to enhance the information logistics related to maintenance support through appropriate information services, both the business process and related technologies need to be linked and aligned seamlessly (see the reasoning in Refs. [26–30]).

Furthermore, there are several existing contributions that deal with the establishment of eMaintenance from a SW architecture perspective. One contribution is the concept of a common database, i.e., an integration at the data level for provision of the needed flexibility for different working areas (e.g. maintenance), as support to strategic, tactic and operational decision making [31]. Another contribution is based on a client-server approach [32]. This client-server architecture is based on two subsystems: (I) a maintenance centre, which is intended to be tailored to the needs of the company through cooperation among the maintenance-related areas; and (II) local maintenance, which implements and evaluates the maintenance system provided by the

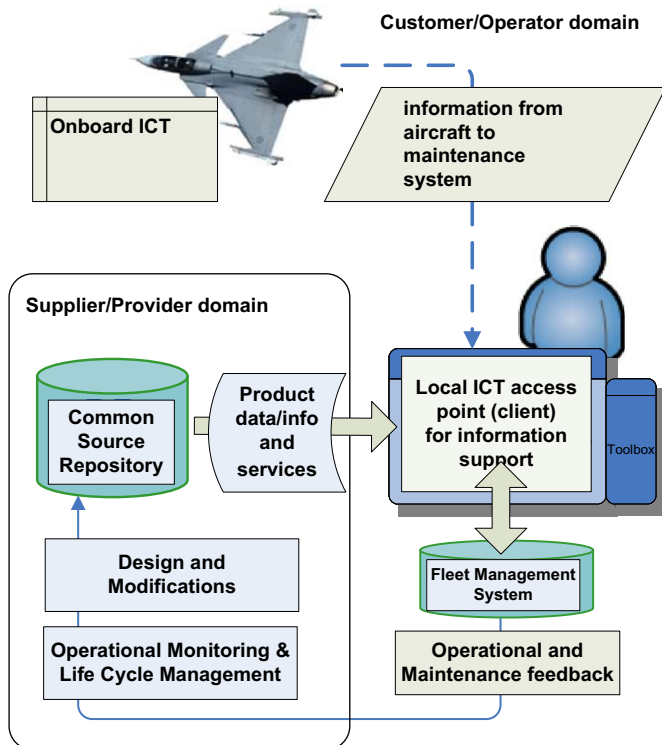


Fig. 4. Content sharing, service provisioning and feedback between supplier/provider and customer/operator domains.

maintenance centre. Another architecture approach relies on a star-like layered architecture [33,34]. One way to realize this star-layered architecture is to use a platform that utilizes the Web services (WS) technology upon the existing maintenance applications to provide function and data integration. However, the Web services technology is only one way to realize a Service-Oriented Architecture. An additional contribution is a proposed eMaintenance Management Framework, which is based on a service-oriented approach for the development of eMaintenance solutions [35].

However, when considering SW-related architecture it can be concluded that SW architecture has evolved from focused on server-computing to client-computing and during the last decades the focus has been changed to distributed computing. Distributed computing has been manifested through various approaches, e.g. client-server, object-oriented, component-based and service-oriented. The latest architectural approach is sometimes called “virtual computing”, which provides an abstraction level of distributed computer resources, e.g. disk spaces, SW applications and processor capacity. Examples of providers that offer support for virtual computing are: HP, Sun, IBM and Microsoft. See Fig. 5.

The latter contribution provided by Karim [35] is described in more detail in the following section.

6. eMaintenance management framework

One way to achieve this is through utilization of an eMMF, see Ref. [35]. As shown in Figs. 7 and 8, the eMMF is a meta-level model through which a range of technologies, methodologies and tools can be clarified and integrated. The main aim of an eMMF is to facilitate maintenance and to ensure its connection to overall business goals. The framework consists of two parts: the eMaintenance Management Model (eMMM) and the eMaintenance

Platform (eMP). The eMMM is a package of roles, processes and repositories required for managing the eMP. The eMP is a Service-Oriented Architecture application, aimed to provide its stakeholders with tailored information for making decisions on the choice of appropriate maintenance activities.

An example of information services related to maintenance actors is the provision of Technical Publication (TP). In the context of Gripen, the TP is structured according to S1000D (see Fig. 6). However, one way to materialize information services related to TP is through two essential eMaintenance services. The first service, i.e., Maintenance Plan Delivery Service (MPDS), aims at delivery of the maintenance plan in a proper format, e.g., Portable Document Format (PDF), eXtensible Markup Language or Standard Generalized Markup Language (SGML). The second service, Maintenance Plan Feedback Service (MPFS), might be aimed at managing feedback from the utilization and support phases. The MPFS aims at the provision of a commenting and reporting capability related to content provided by MPDS on issues raised on data modules or publication modules during the verification process and the in-service phase of Gripen.

The eMP is divided into three levels: Specification, Design, and Implementation (see Fig. 8). The Specification level contains all the information necessary for conducting maintenance activities. It describes the identified requirements, which are realized in various ways using notations such as text and models. The Design level identifies design components that are needed to fulfil the requirements specified in the Specification level. This level is realized as service components. Finally, the Implementation level materialises the service components specified in the Design level into Web services and business processes orchestrated for specific needs. Each level consists of one or several groups of components. All components in the Specification level provide a platform for creating other components in the Specification level or they get directly realized on the Design and/or Implementation levels [37].

The eMP provides artefacts for supporting the maintenance process, e.g. situation-adapted information and services for performance support [4,5]. Further, SOA can be exploited

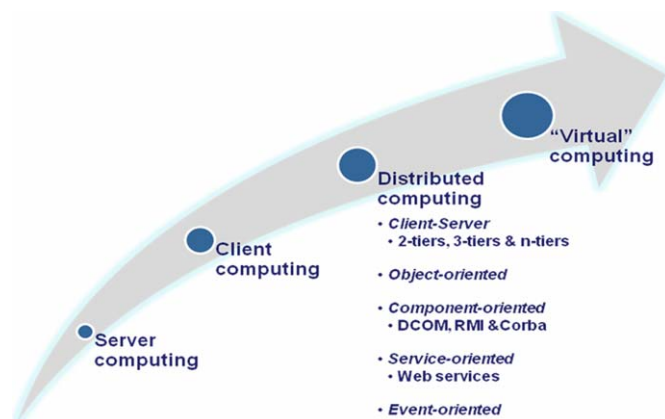


Fig. 5. The architecture maturity [35].

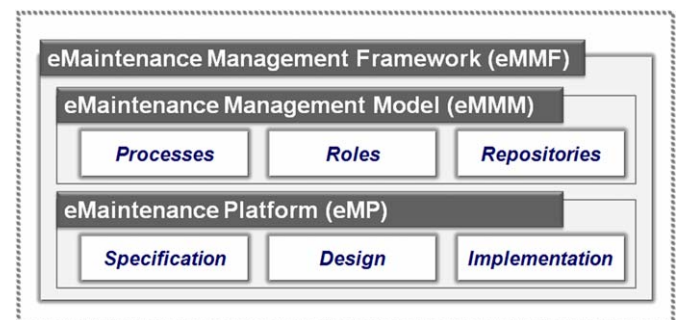


Fig. 7. Illustrating the components in the eMMF [37].

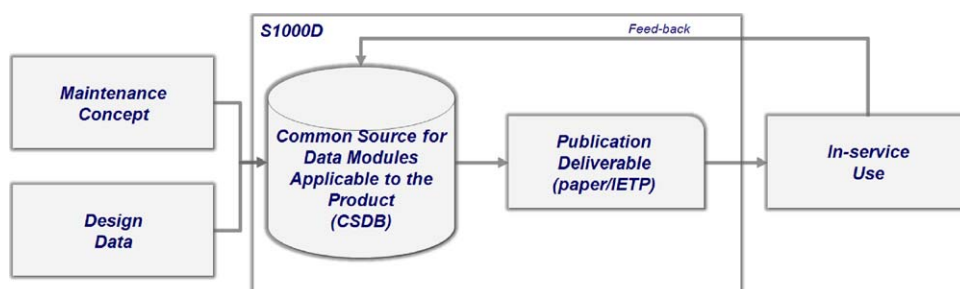


Fig. 6. The scope of s1000D [36].

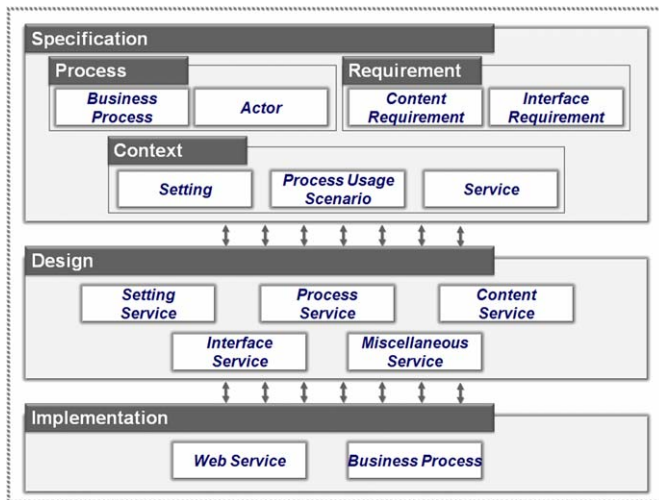


Fig. 8. Illustrating the substances in eMP [37].

in eMP. Hence, the eMP can be utilized to implement an eMP in organizations that provide maintenance of complex technical systems. The eMP is also most likely applicable in any organization providing maintenance, since it is based on a generic overall maintenance process.

The Specification level contains all the information necessary for conducting maintenance activities. It consists of three components: Process, Requirements and Context. The Process component lists and describes the business processes and actors involved in them. It consists of two parts: Business Process and Actor. The Context component lists and describes various backgrounds and their interrelated conditions in which maintenance activities take place. It consists of three components: Setting, Process Usage Scenario and Service. The Requirement component lists and describes all the functional and non-functional requirements on the information needed for a specific maintenance activity and the interface through which this information should be presented. It consists of two components: Content Requirement and Interface Requirement.

The Design level transforms most of the specification components into SOA components. The role of the Setting Service component is to design choreography for a certain set of services. The role of the Process Service component is to design a composition of miscellaneous services orchestrated for serving a particular process. The role of the Content Service component is to design services that implement logic for interaction with different data sources, such as existing systems or legacy systems. The role of the Interface Service component is to design services that handle/manage/wrap/adapt interface to other devices such as a portable handheld computer. For the exposition of services several technologies might be utilized, e.g. Distributed Component Object Model (DCOM), Remote Method Invocation (RMI) and Web services. If it is required to provide accessibility to the services via Hyper-Text Transfer Protocol (HTTP) over internet and data exchange through XML, Web services were selected as an appropriate technology. However, the service interfaces can be designed based on the Web services, which mean that using Simple Object Access Protocol (SOAP) envelope for information exchange. This implementation of services also provides the capability to expose a description of the signature of inherent public web methods, in accordance with the Web Services Description Language (WSDL) standard.

The Implementation level materializes the SOA components into the Web services and business processes orchestrated for the needs at hand. The Web Service component implements the SOA

components in the design level as Web Services. Web services technology is used to materialise these components, since they provide a generic interface for encapsulating logics and can be utilized as an intermediate layer to access legacy systems. The Business Process component corresponds to business processes as orchestrated for the needs at hand and delivered to the service consumer. In the context of eMaintenance, there is a need for both predetermined business processes and reactive business processes. To increase the effectiveness and efficiency in the implementation of services it is ought to be supported by a proper Interactive Development Environment (IDE) with capability to provide functionality necessary for team development, e.g. Microsoft .NET platform and IBM Web Sphere software.

7. Conclusions

The positive impact of eMaintenance is to be expected on two levels. One is the 'maintenance micro-level', where eMaintenance will serve as a performance support that facilitates hands-on execution of maintenance tasks by technicians, mechanics and support engineers, by providing a reduced number of interfaces to information sources, improved fault diagnosis, knowledge sharing and automated or facilitated procedures for technical administration. The other level is the higher 'maintenance macro-level', where eMaintenance will support managerial maintenance planning, preparation and assessment, enabling information-driven maintenance and support processes, and fleet-efficiency, e.g. aircraft maintenance programme evaluation, elimination of redundant information and operational monitoring. The impact of an eMaintenance implementation on aviation maintenance is of considerable potential as it enables a more efficient use of existing digital product information and design data over the whole life cycle.

There is also the challenge of going from a product-oriented to a service-oriented business strategy. A product-oriented strategy relies on a transaction/exchange marketing, while service-oriented emphasizes the relation between provider and customer. A service-oriented business strategy also demands the harmonization of support processes, such as the maintenance process, to the business's core process. The business's core process ought to be adaptable to the changes in the customers' value-generating process, and the business's supporting processes need to be adaptable to the core process. This change is facilitated by the proposed eMaintenance Management Framework and its SOA approach.

Since an eMaintenance platform's major purpose is to support the maintenance process, the components of the platform need to be considered from a service-oriented perspective in order to increase its adaptability to different parts of the maintenance and support process in different organizations. Hence, eMaintenance provides a platform for smart content sharing between information provider and information consumer. In this paper, the term-pair provider/consumer is used to emphasize that the ICT support to the maintenance process should adapt a process-centric approach, supporting efficient maintenance and support solutions. In a supplier/customer approach, the definition of the organizational boundary is important, which is not necessarily an issue of importance for delivery of maintenance information.

The maintenance process has to interact with a number of correlated processes in an enterprise, e.g. business, operation, information, support and logistics. Hence, a service-oriented eMaintenance platform should provide services that can be incorporated and orchestrated to the flow of these processes. While the requirements on the maintenance process vary over

time, the architecture of eMaintenance should be constructed to provide applicability and effectiveness.

With applicability and effectiveness as the two major requirements, it is clear that a Service-Oriented Architecture approach is a prerequisite for being able to design a solution which will provide applicable and effective eMaintenance services. Furthermore, it should be noted that a Service-Oriented Infrastructure is a prerequisite for implementing SOA. It is important to be aware of that Web services, and their related technologies, e.g. WS-BPEL, WS-Transaction and WS-Coordination, are not synonymous with SOA. SOA is a way of thinking that approaches the architectural issues and WS is a technology that is designed to facilitate the implementation of SOA.

We believe, based on our study of the effects of eMaintenance, that the eMaintenance platform, mainly through its improvement of different parts of the maintenance process, impacts on the dependability, safety and life-support costs of critical systems. Hence, our conclusion is that an eMaintenance Platform structured, designed and implemented from a service-oriented perspective with focus on the business process, increases the ability to fulfil requirements such as context-awareness, situation-awareness, seamless information integration between processes (e.g. operation and maintenance), improved knowledge-sharing, flexibility, extensibility and cost-reduction. Further, we can conclude that SOA as an approach and Web services as a technology can be utilized to implement service-oriented information services related to maintenance and its actors. Loosely connected maintenance services can, relatively, easily be re-composed and orchestrated to the new process, situation and context to better fulfil different stakeholders' requirements.

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